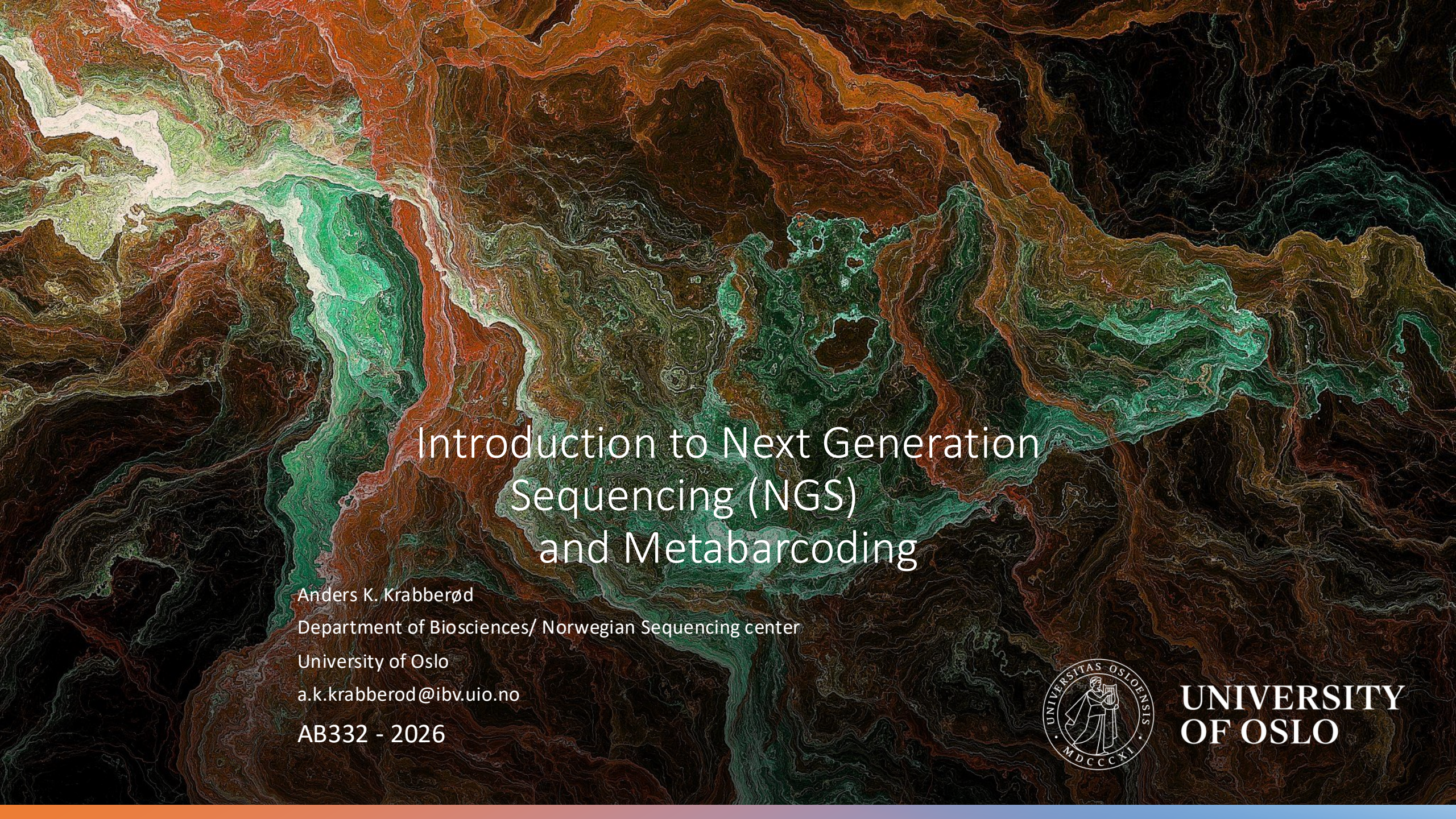


Introduction to Next Generation Sequencing (NGS) and Metabarcoding

Anders K. Krabberød
Department of Biosciences/ Norwegian Sequencing center
University of Oslo
a.k.krabberod@ibv.uio.no

AB332 - 2026

UNIVERSITY OF OSLO

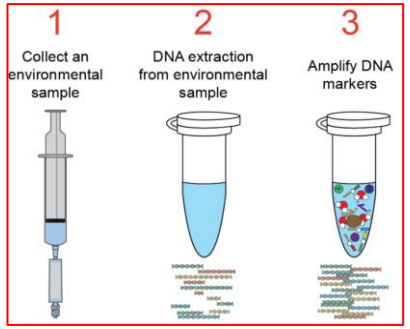
Some important terms

- Metabarcoding – The use of DNA markers to analyze biodiversity and species identification from environmental samples.
- Amplicons - DNA fragments generated through polymerase chain reaction (PCR)
- OTUs (and ASVs) - Groupings of closely related sequences used as proxies for species (Operational Taxonomic Unit, Amplicon Sequence Variants)
- High-throughput Sequencing (HTS) - A broad range of technologies that enable the rapid sequencing of large volumes of DNA or RNA
- Next Generation Sequencing (NGS) - A specific type of HTS technology that allows for massively parallel sequencing of DNA (or RNA)
- Third-generation Sequencing - A newer generation of sequencing technologies (e.g., PacBio, Oxford Nanopore) that read long, single molecules of DNA or RNA in real-time

Metabarcoding

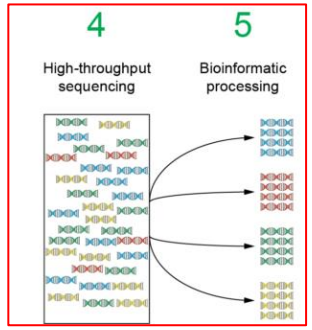
Fieldwork/labwork/
DNA extraction etc

Not covered in this lecture



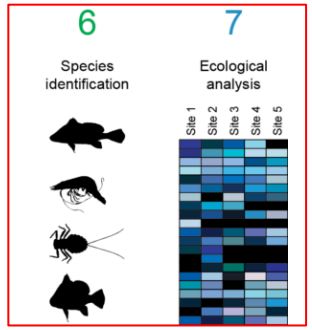
Sequencing /
Bioinformatics

Tuesday

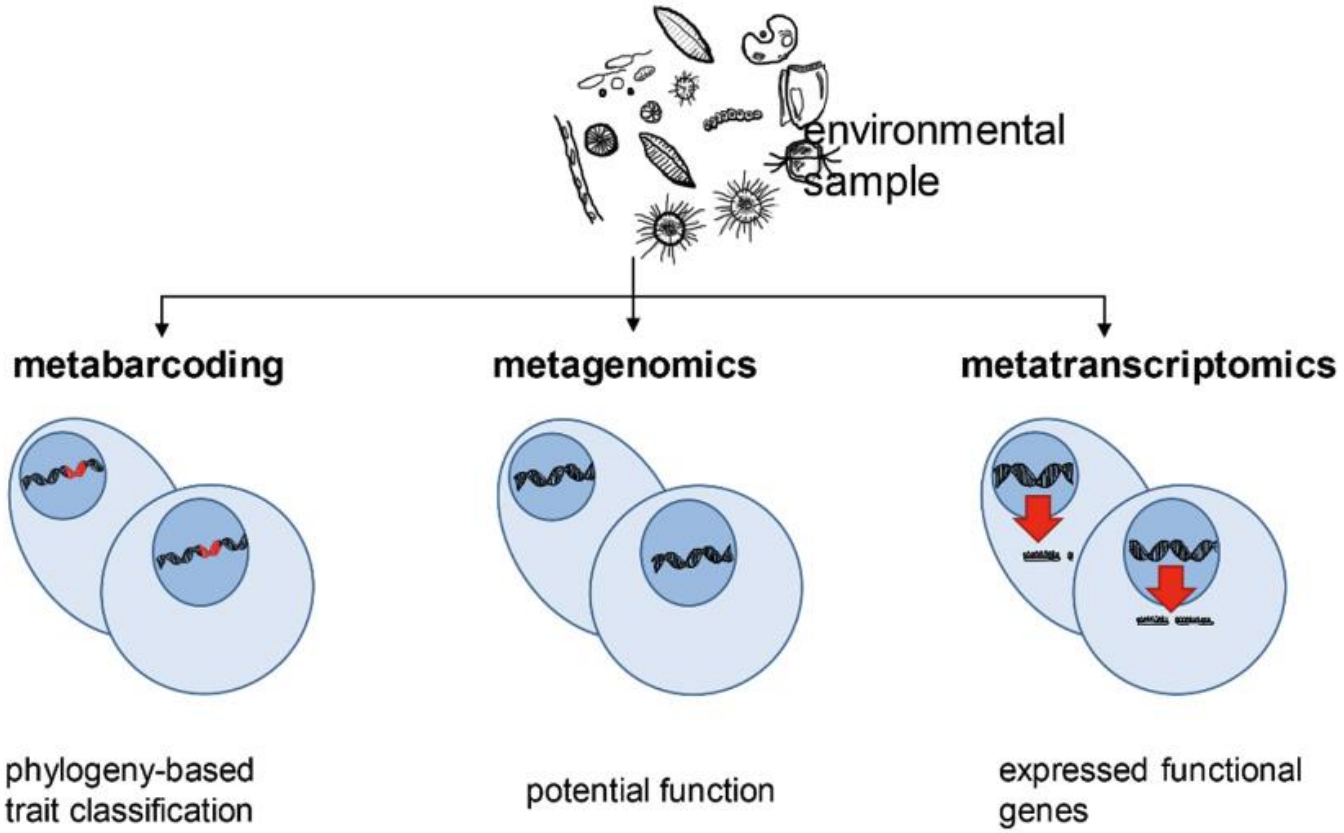


Ecological
analysis

Computer Labs
Tuesday and Wednesday



Abdelfattah et al.,2018

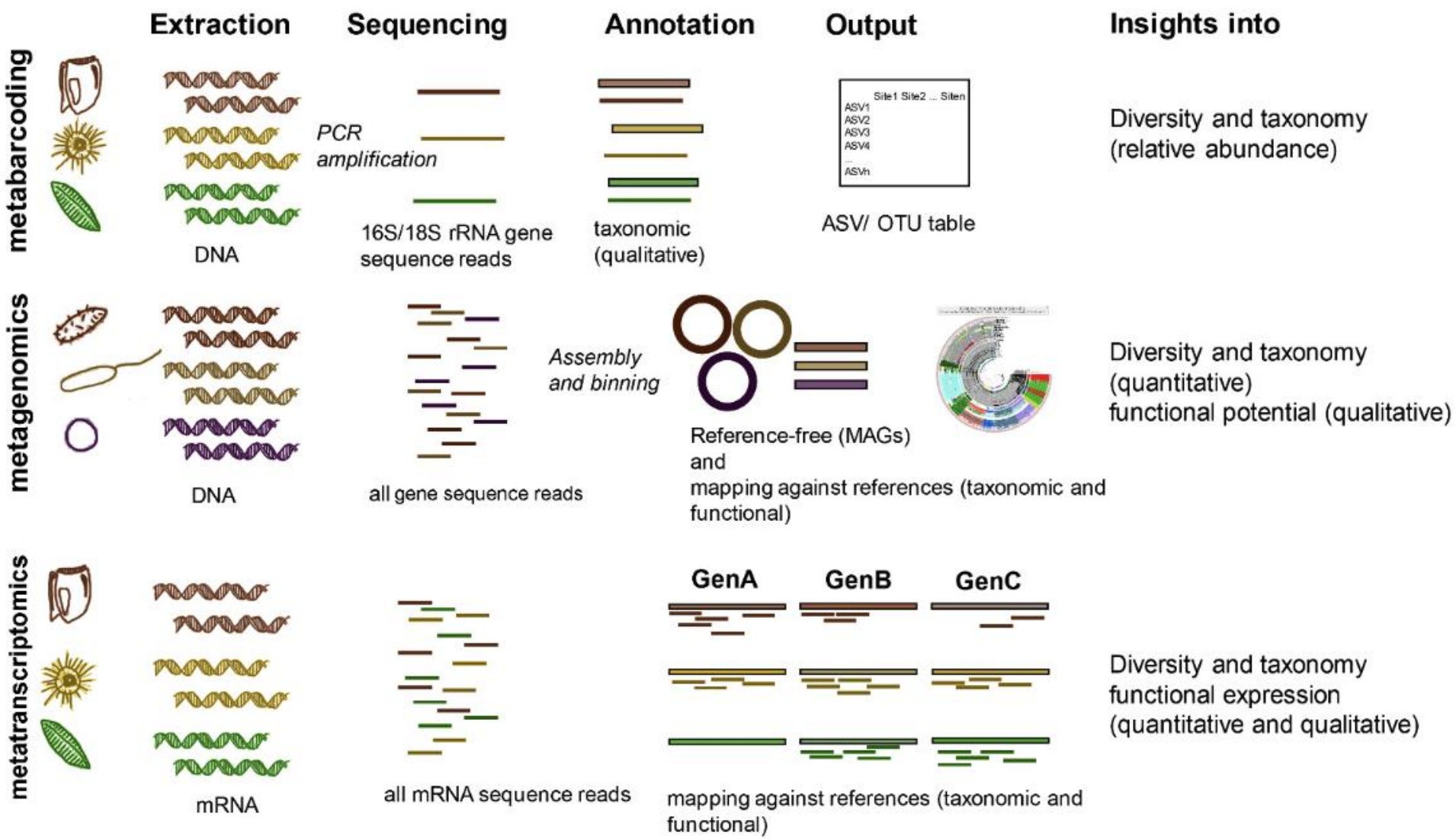


Who is present?

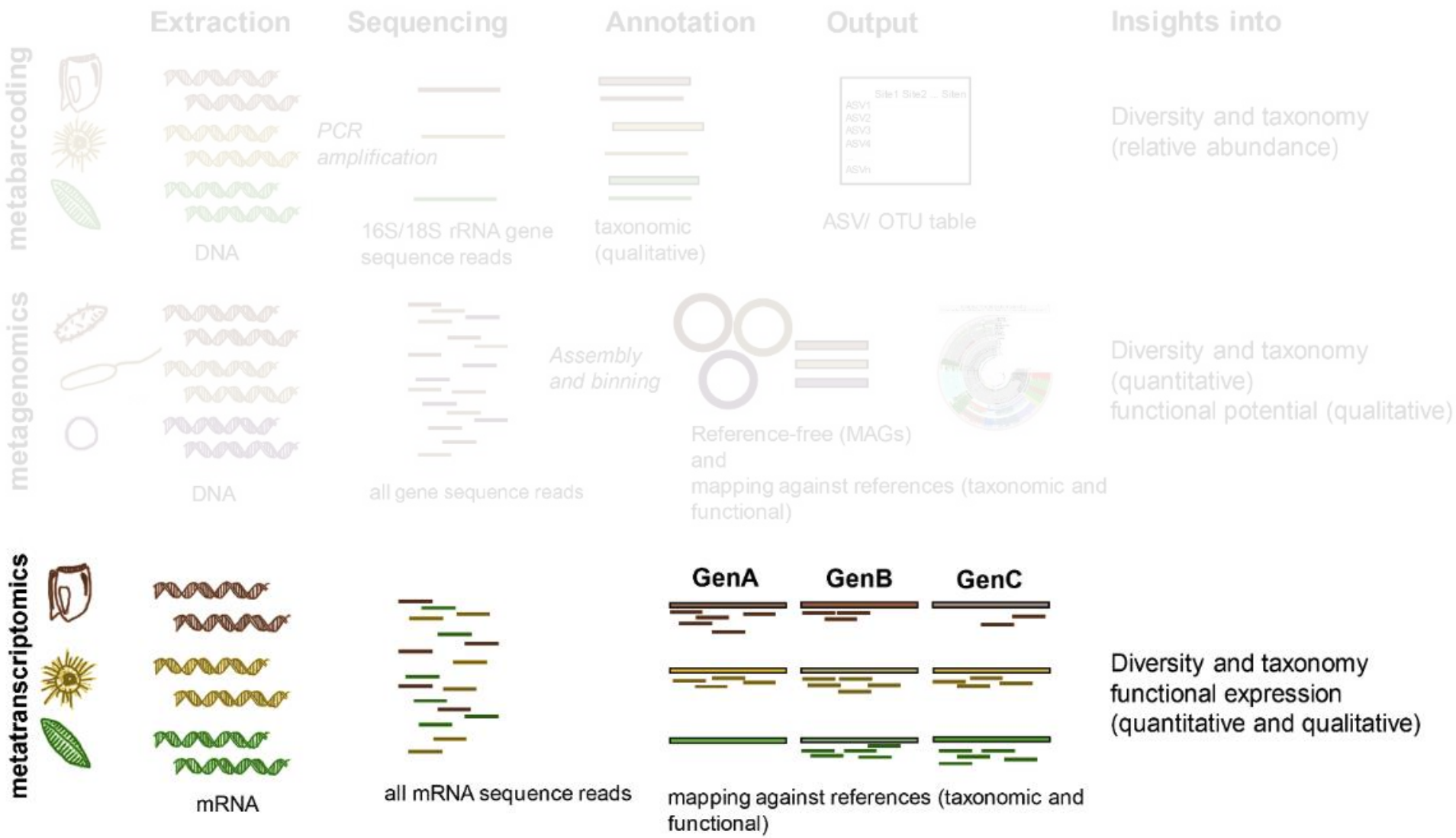
Who, and potential functionality

Who is active, and which genes?

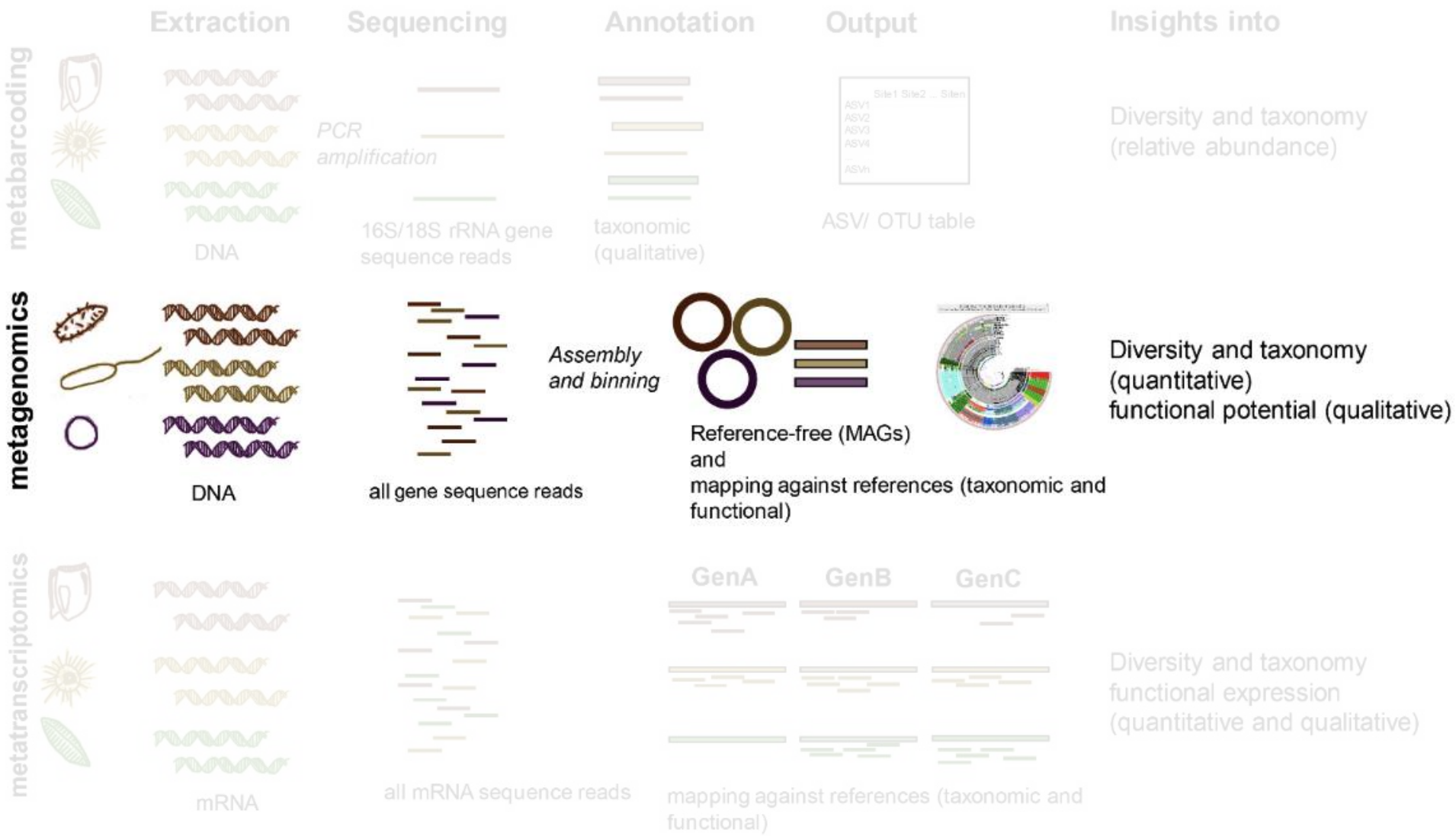
Hoerstmann et al 2022



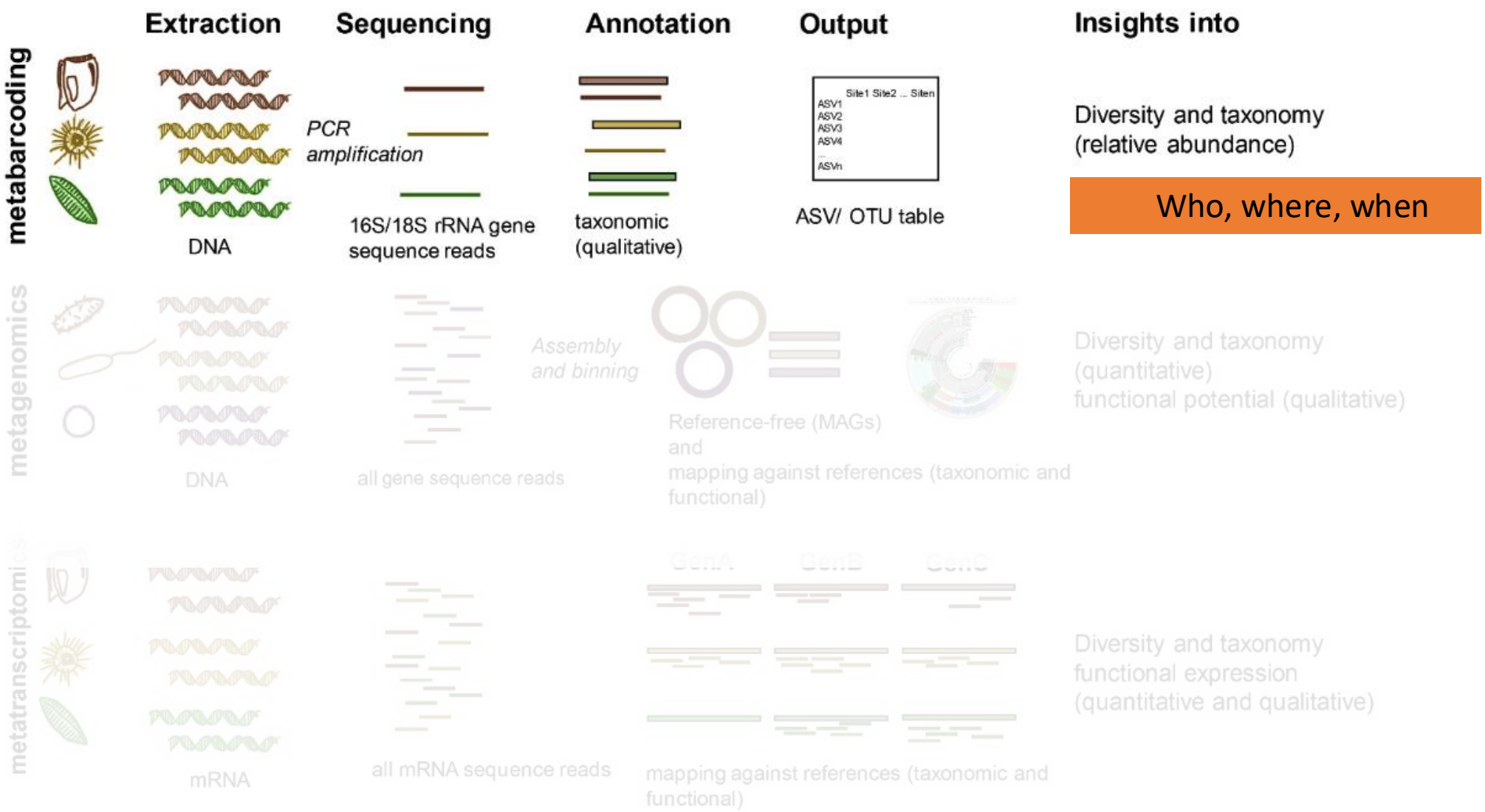
Hoerstmann et al 2022



Hoerstmann et al 2022



Hoerstmann et al 2022



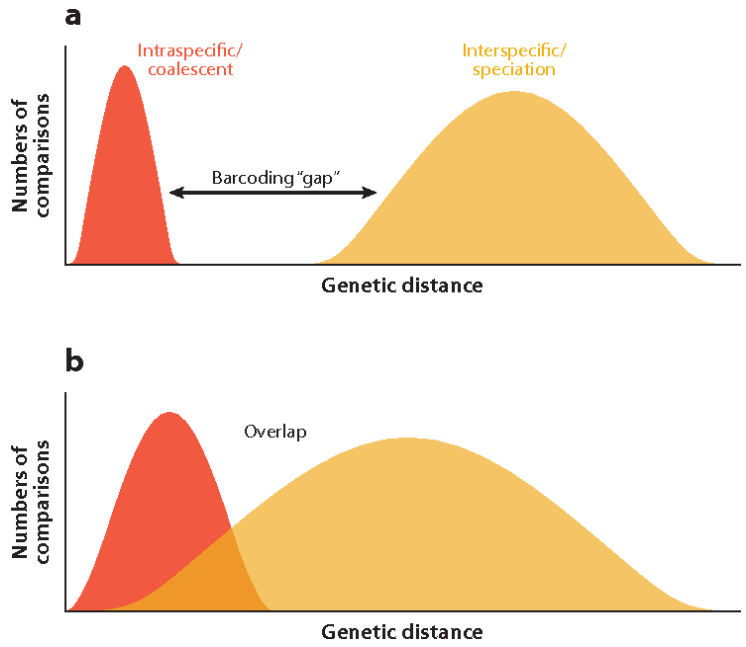
Hoerstmann et al 2022

Metabarcoding

- Typical research questions:
  - Who is present, where, and when?
  - Richness: How many taxa/species/OTUs (alpha/gamma diversity)?
    - **OTU:** Operational taxonomic unit -> the group of organisms currently being studied (Sokal & Sneath 1957), a modern take would be that OTUs are pragmatic proxies for "species".
  - Compositional differences (beta diversity)?
  - Which processes and drivers are shaping the communities?
  - Co-occurrence patterns (possible interactions)

Markers in DNA metabarcoding

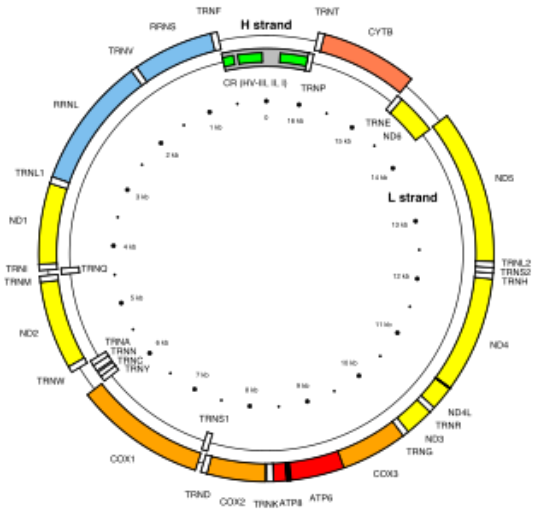
- The ideal marker should:
  - Have primer sites that are shared by all target organisms
  - Be easy to amplify in PCR
  - Be of appropriate length for efficient *amplification* and *sequencing*
  - Be of similar length for all target organisms
  - No intragenomic variation (i.e. no paralogs)
  - Similar number of copies
  - Have high interspecific variation
  - Have low intraspecific variation
  - Be possible to align (not always required)
- Likely no marker will meet all these requirements!

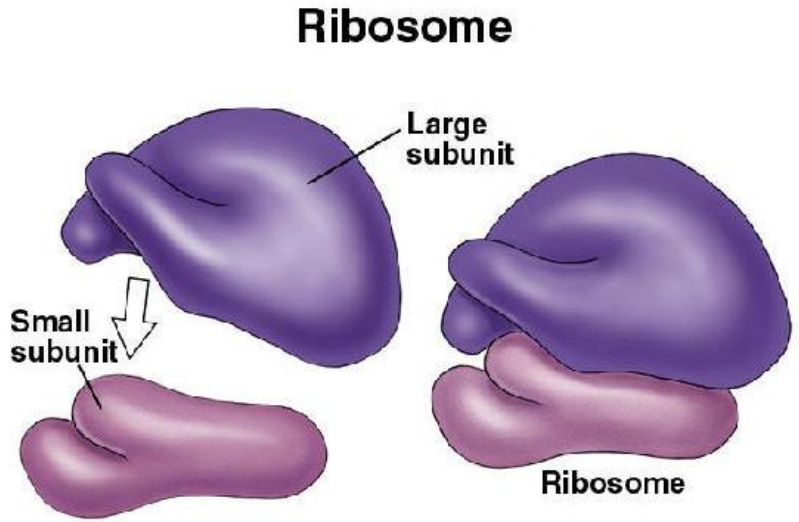


Markers used in DNA metabarcoding

- Standard markers (<500 bp):
  - 18S: Eukaryotes
  - 16S: Bacteria/archaea
  - ITS: Fungi & plants
  - COI: Metazoa
  - *Rbc*L: Plants
  - *trn*L: Plants

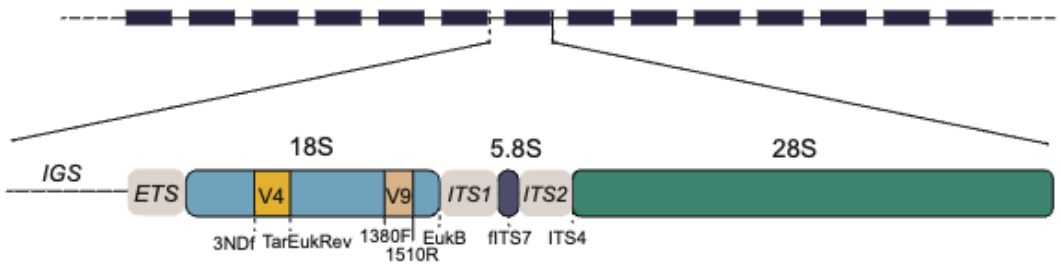






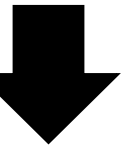
Ribosomal operon

- Often shortened rRNA or rDNA
- 16S – 5S - 23S in prokaryotes
- 18S – 5.8S -28S in eukaryotes
  - S stands for Svedberg units; a unit of molecular size determined by centrifugation.
- Present across the Tree of Life!
- The full length is in the range of 5000-7000 (but with a lot of variation)
- Typically for Illumina sequencing a region of 300-450bp is used (e.g. V4).



How conserved/variable are the marker?

- 18S (and 16S): (Relatively) Low variability, low intraspecific variation, low interspecific variation
- ITS: High variability, high intraspecific variation, high ‘interspecific’ variation

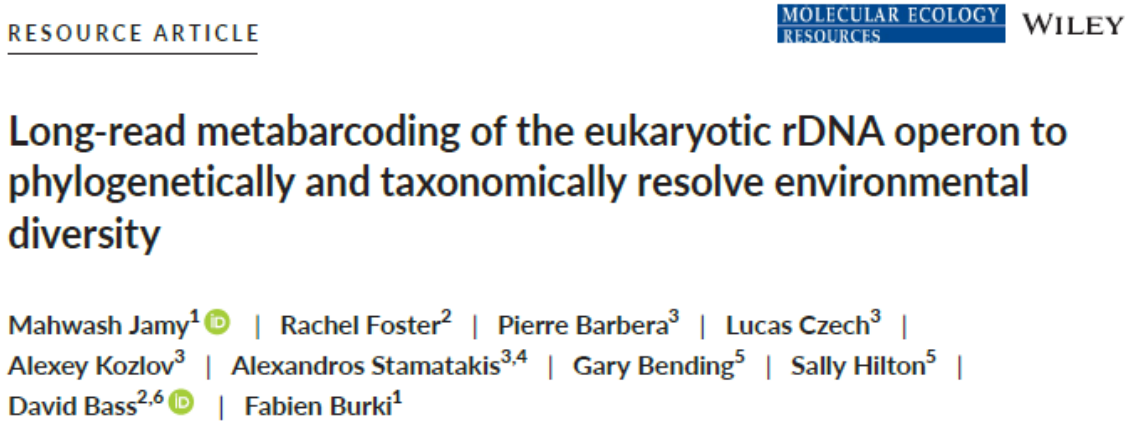


- Impact how the bioinformatics analyses should be conducted

Long-read metabarcoding

- Sequencing the full operon is possible with the development of sequencing technologies
- Stronger phylogenetic signal

- Comes with extra challenges
  - Harder to amplify longer regions
  - More chimeric sequences
  - Lower sequencing depth



RESOURCE ARTICLE

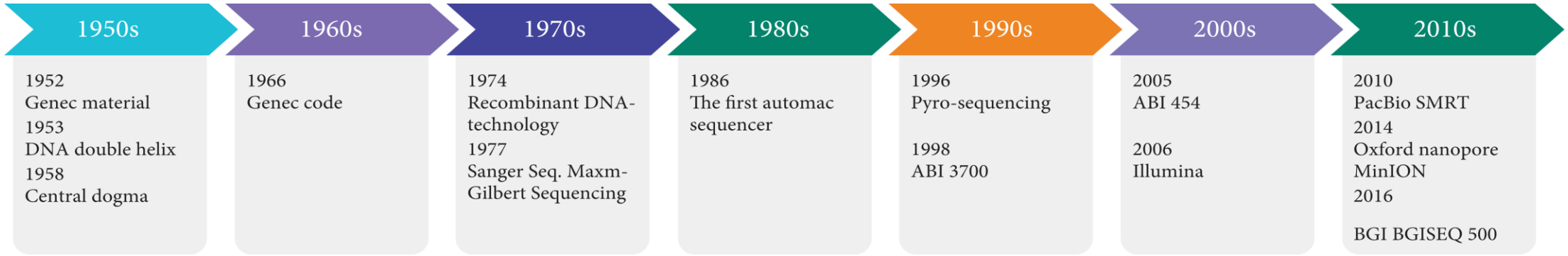
MOLECULAR ECOLOGY RESOURCES WILEY

Long-read metabarcoding of the eukaryotic rDNA operon to phylogenetically and taxonomically resolve environmental diversity

Mahwash Jamy[1] | Rachel Foster[2] | Pierre Barbera[3] | Lucas Czech[3] | Alexey Kozlov[3] | Alexandros Stamatakis[3,4] | Gary Bending[5] | Sally Hilton[5] | David Bass[2,6] | Fabien Burki[1]

A short history of Sequencing

- Sequencing: determining the order of nucleotide bases in a string of DNA (or RNA)
- The development started in the 1950's after Watson and Crick described the structure of DNA
- The early methods were cumbersome (and dangerous) using radioactive material and adding individual nucleotides to a reaction one by one.
- The last few years the development has been phenomenal!



A short history of Sequencing

- F. Sanger et al. 1977
  - Short fragments
  - 15-200 nucleotides
  - Sloooooow process

Proc. Natl. Acad. Sci. USA
Vol. 74, No. 12, pp. 5463–5467, December 1977
Biochemistry

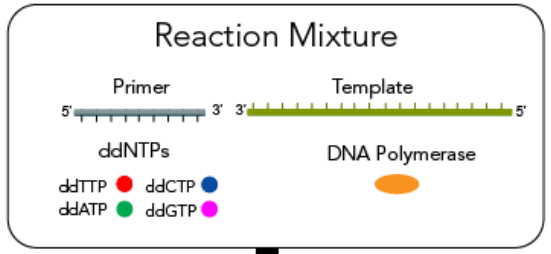
DNA sequencing with chain-terminating inhibitors

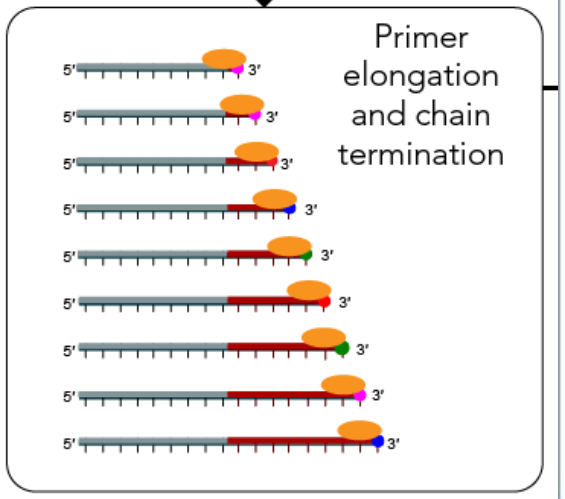
(DNA polymerase/nucleotide sequences/bacteriophage ϕX174)

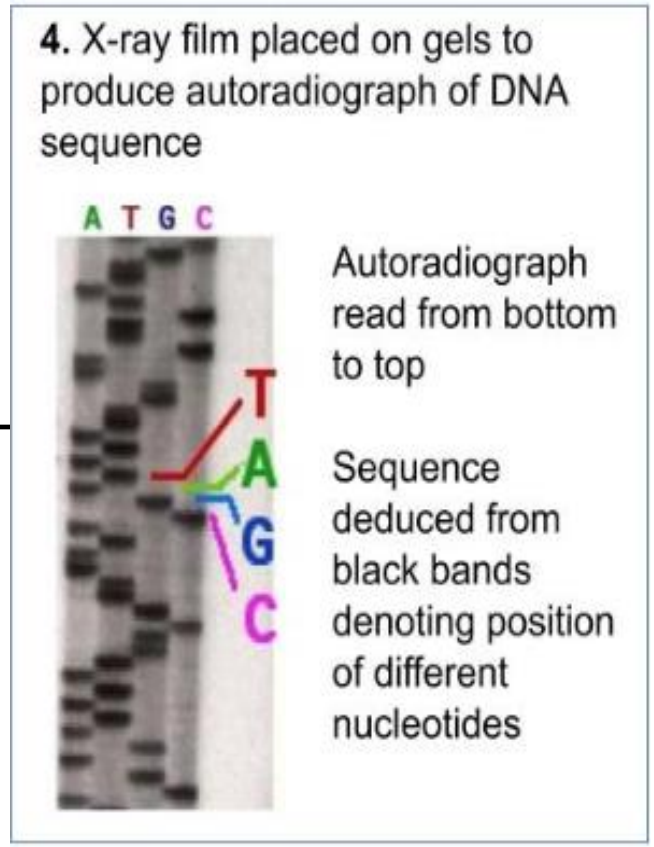
F. SANGER, S. NICKLEN, AND A. R. COULSON

Medical Research Council Laboratory of Molecular Biology, Cambridge CB2 2QH, England

Contributed by F. Sanger, October 3, 1977

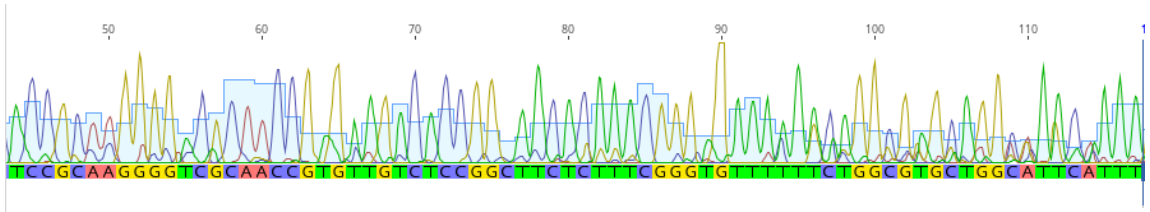


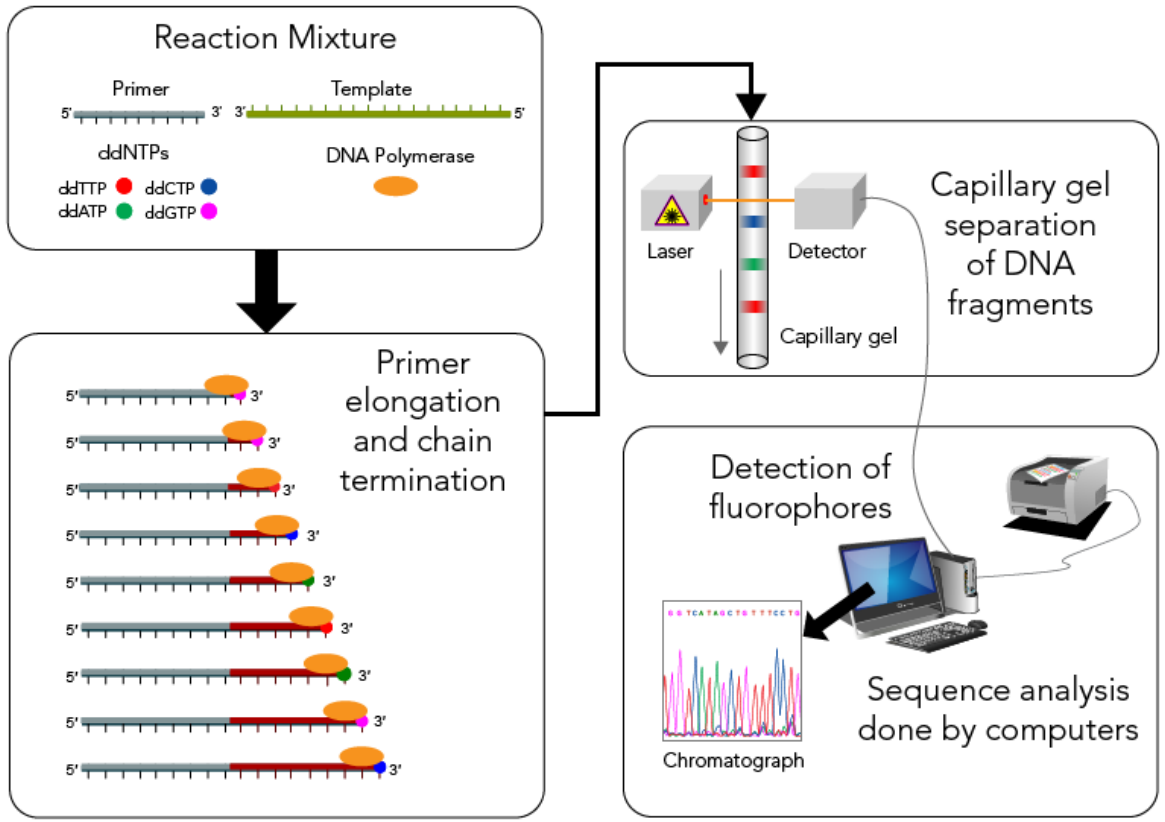




A short history of Sequencing

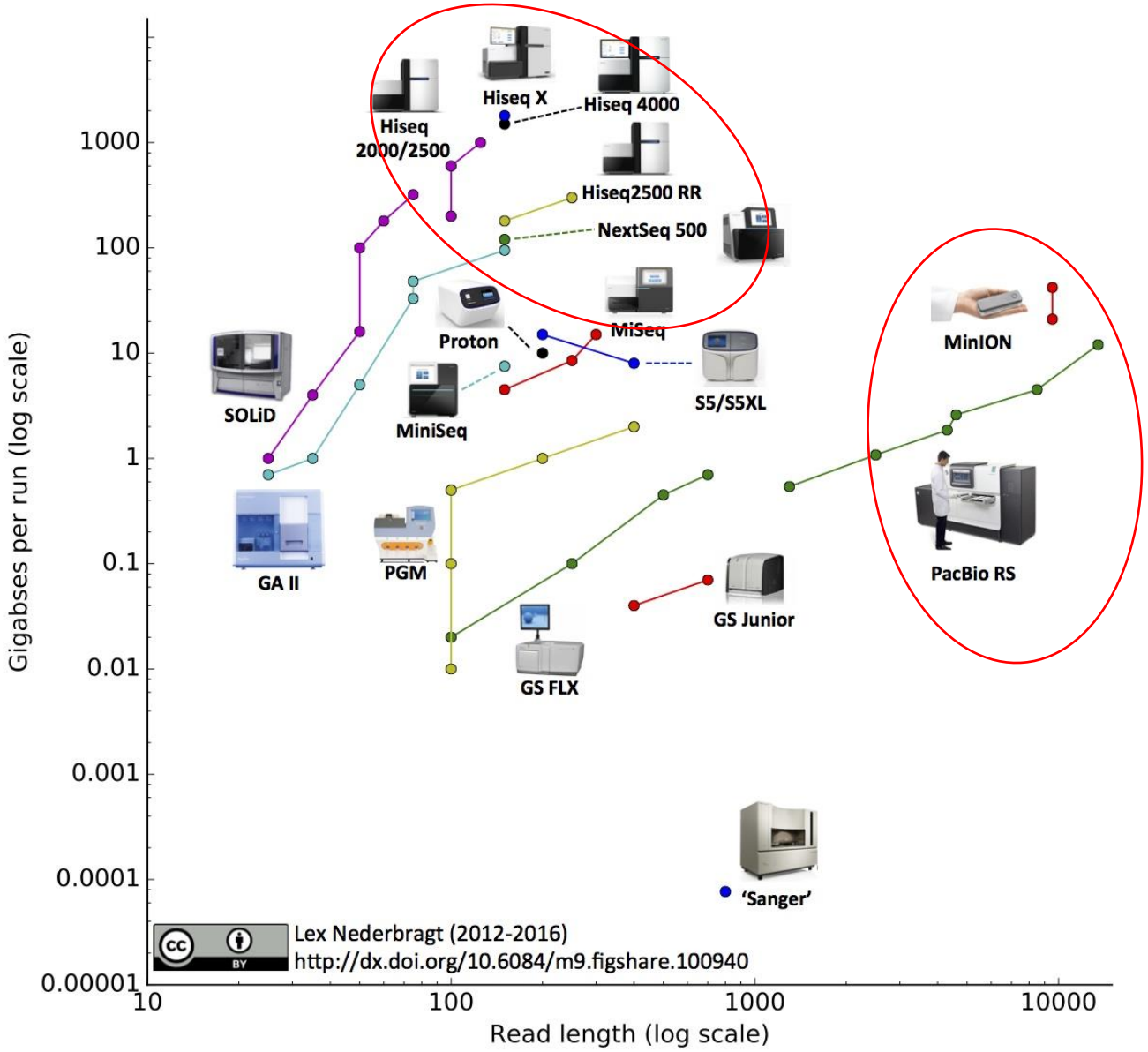
- F. Sanger et al. 1977
  - Short fragments
  - 15-200 nucleotides
  - Sloooooow process
- Applied Biosystems automating the process in the late 80's early 90' with capillary electrophoresis, fluorescent dyes, and lasers.
- "First generation sequencing"
  - 500—1000 nucleotides
  - Slow, but much faster than manual
  - The main technique for the human genome project
  - Sequencing 3 gigabases took 10 years
  - Still used, since it is very high quality and cheap (if you only want to look at a handful of sequences)





High Throughput Sequencing (HTS)

- Early 2000's and onwards
- High Throughput Sequencing (HTS) is a collective term
  - Next generation sequencing (NGS)
    - Short reads, (100-300bp)
    - but generates a huge amount of reads (in the billions)
    - 454 Roche
    - **Illumina (HiSeq, MiSeq, NovaSeq, NextSeq)**
    - Ion torrent
  - Third generation sequencing
    - Longer reads, (1000-100kbp)
    - not so many as NGS, but still in the 100k or millions
    - Oxford Nanopore (MinIon, GridIon,PromethIon, etc)
    - **PacBio (Sequel, Revio)**


Gigabses per run (log scale)
1000
100
10
1
0.1
0.01
0.001
0.0001
0.00001
10
100
1000
10000
Read length (log scale)
Hiseq 2000/2500
Hiseq X
Hiseq 4000
Hiseq2500 RR
NextSeq 500
Proton
MiSeq
SOLiD
MiniSeq
S5/S5XL
MinION
GA II
PGM
PacBio RS
GS Junior
GS FLX
'Sanger'
Lex Nederbragt (2012-2016)
http://dx.doi.org/10.6084/m9.figshare.100940

Illumina

- “Sequencing by synthesis”
- Short fragments
  - 150-300 bp in pairs
- Low error rate (0.1% - 0.5%)
- MiSeq output (2*300bp):
  - 25 million reads (15Gb)
- NextSeq (2*300 bp):
  - 1.2 billion reads (360Gb)
- NovaSeq 6000
  - 20 billion reads (6Tb)
- Other platforms exist



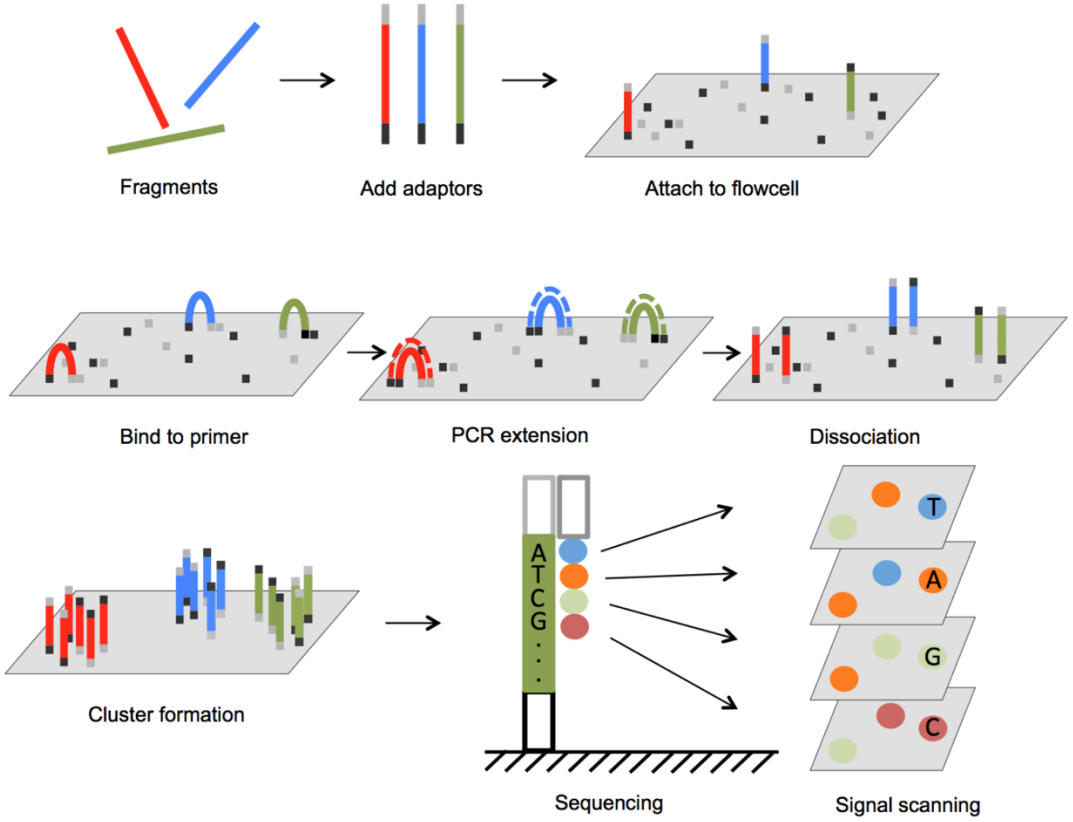


Figure 1: Principle of the illumina sequencing by synthesis (SBS) technology (Lu et al., 2016)

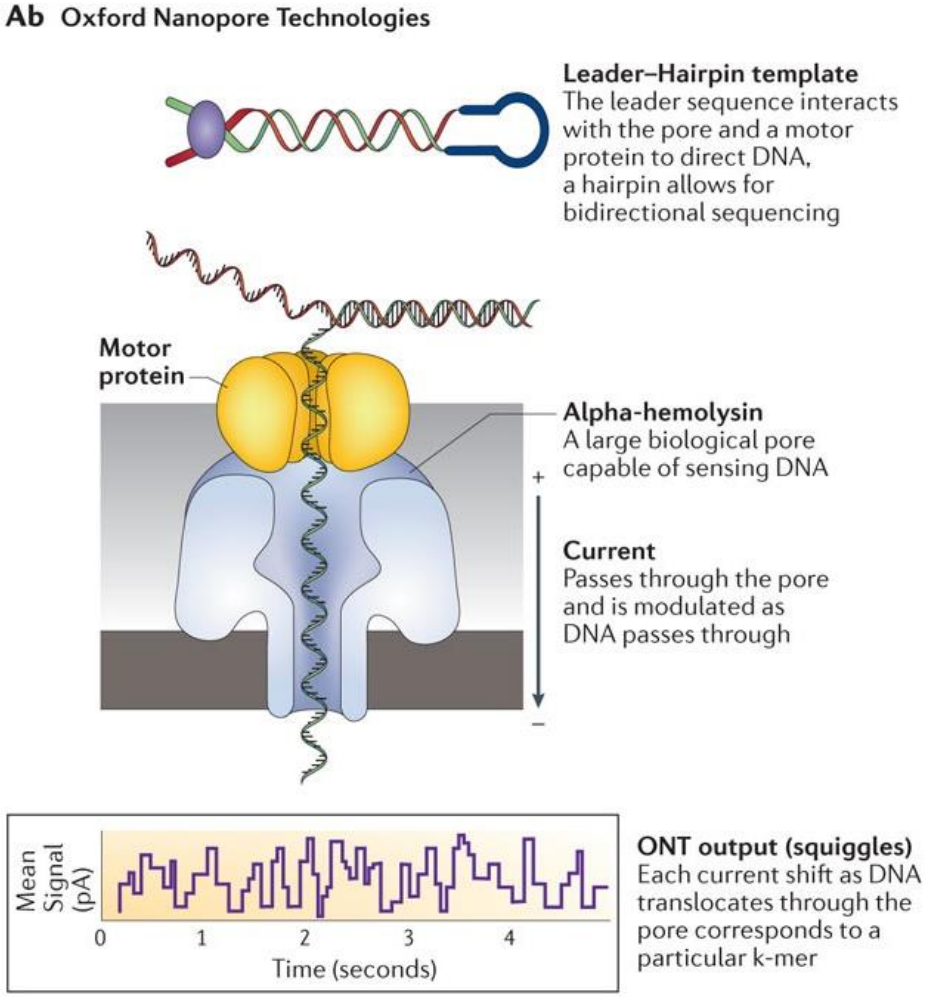
https://www.youtube.com/watch?v=fCd6B5HRaZ8&t=1s

Goodwin, et al 2016

Oxford Nanopore

- Long to very long reads (10kb -100kb)
- Higher error rate, but it is improving
- Lower output than Illumina
  - MinION (50Gb)
  - PromethION (290Gb)
- Realtime sequencing
- Portable





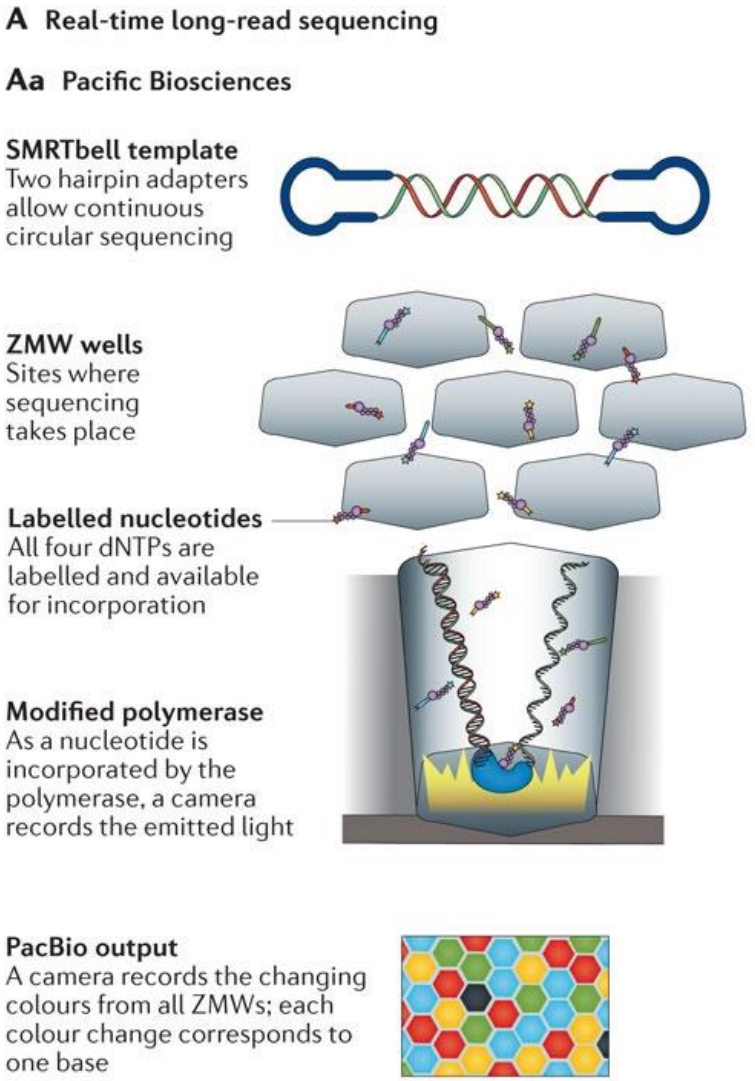
https://www.youtube.com/watch?v=RcP85JHLmnI

Goodwin, et al 2016

PacBio

- SMRT-sequencing
  - Single-molecule Real Time
- Long reads (~15kb)
- Low error rate (0.1%)
- High output
  - Theoretical output:
  - Sequel II (up to 8M reads, 120 Gb)
  - Revio (up to 23M reads, 3Tb)
  - (Real output is ~75% of this)





https://www.youtube.com/watch?v=NHCJ8PtYCFc

Goodwin, et al 2016

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)

- These are the main steps when working with HTS data
- The details will vary depending on the sequencing technology used, the community under study, and the scientific answers being asked
- Some pipelines are built to do all steps for you automatically (Qiime, LotuS)
- Or you need to pick the relevant tool for your data

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Initial cleaning and quality controls

Removal of non-target organisms

Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)

Post processing steps

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

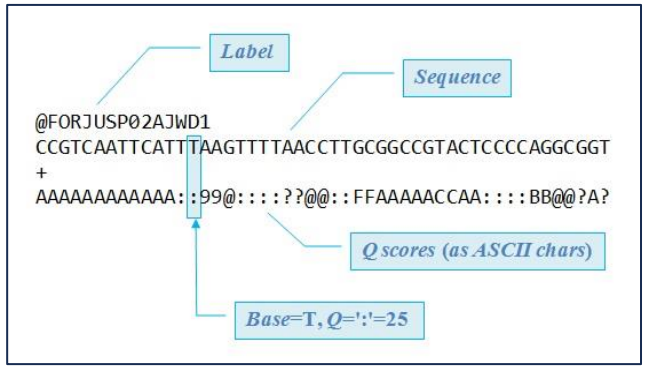
OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)

The *fastq* format



The quality is in phred score

- 1-60, coded in ASCII characters
- 20 is 99% accuracy, 30 is 99.9%
- For a modern interpretation: https://fastqe.com/

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

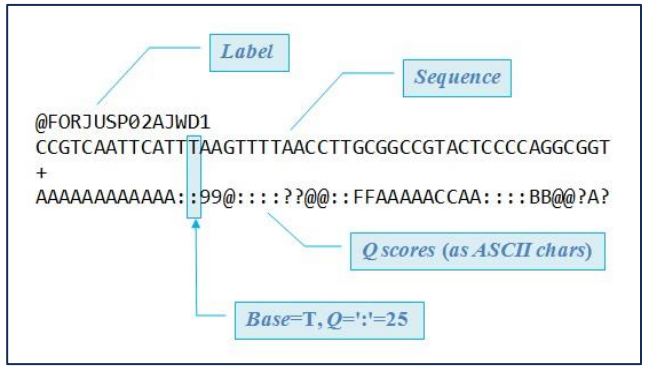
OTU modifications

Positive negatives

Singleton removal

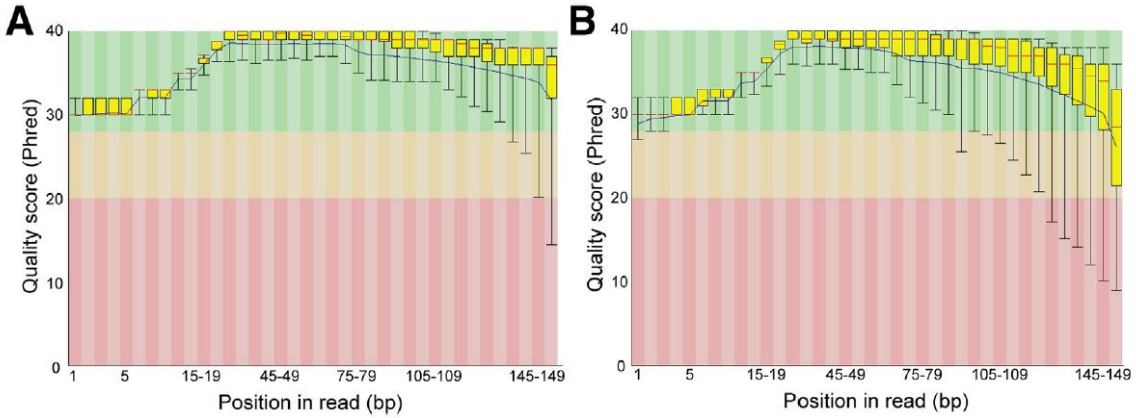
Data transformation (e.g rarification)

The *fastq* format



The quality is in phred score

- 1-60, coded in ASCII characters
- 20 is 99% accuracy, 30 is 99.9%
- For a modern interpretation: https://fastqe.com/



Remove

- Poor sequence quality
- Long/short sequences

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

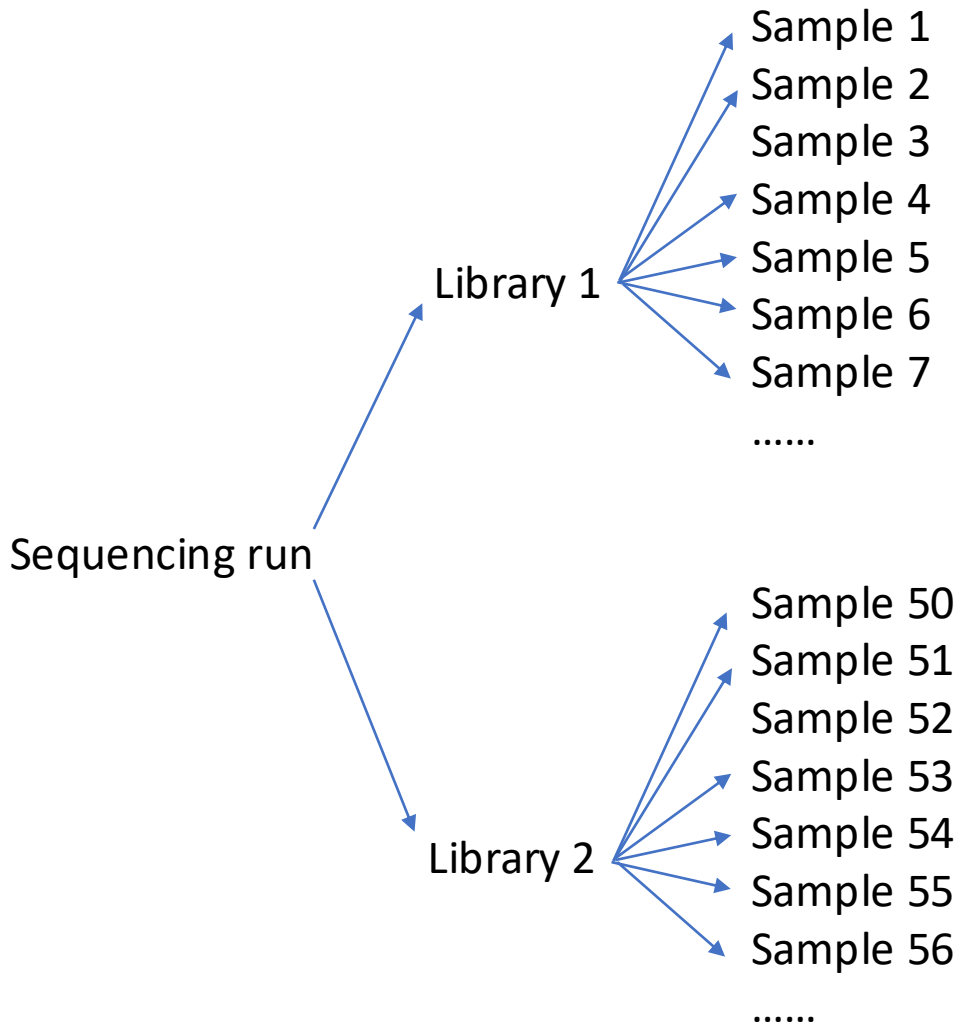
Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)



Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)



(k-mer based analyses)

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

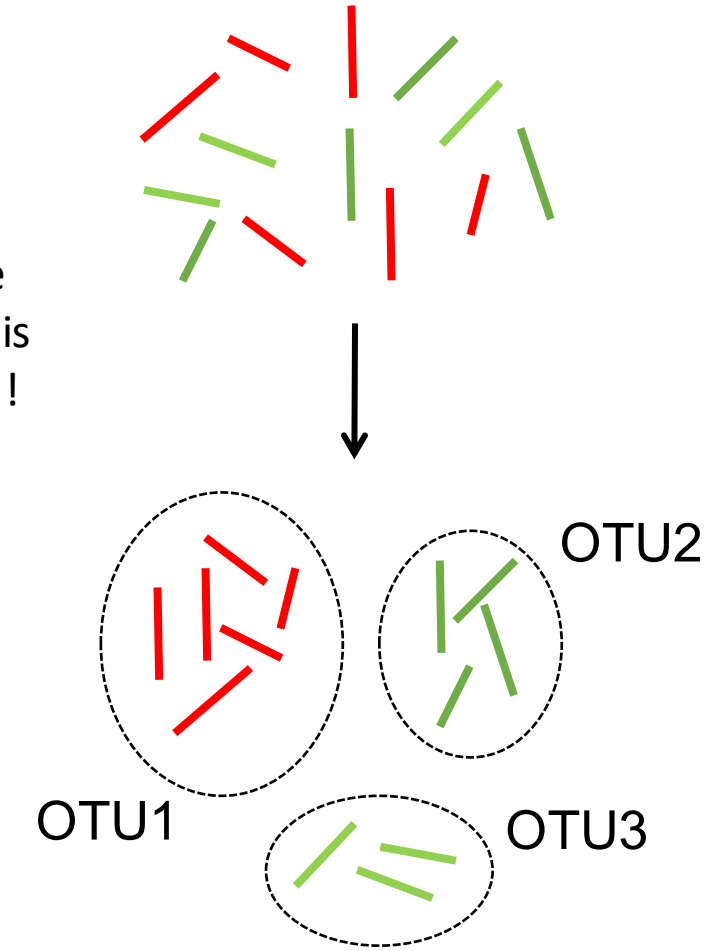
OTU modifications

Positive negatives

Singleton removal

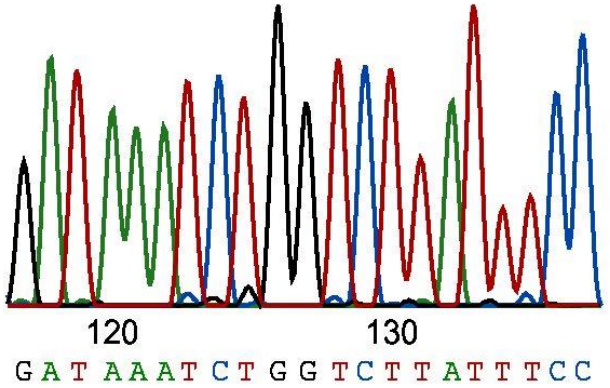
Data transformation (e.g rarification)

More on this later!!

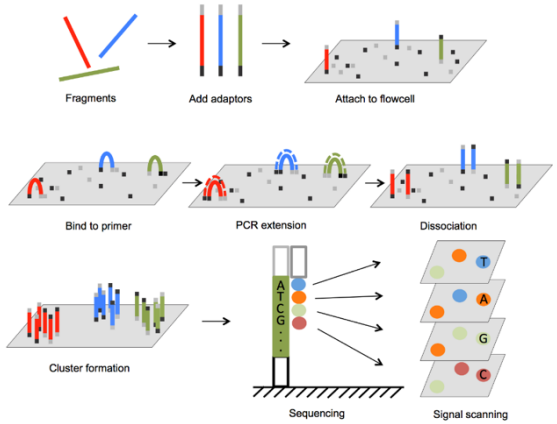


PCR-induced errors

- PCR mutations: polymerase enzymes introduce erroneous nucleotides now and then, even those enzymes with proof-reading activity
  - Dependent on the technology whether these become «visible»
  - In classic (direct) Sanger sequencing such errors become «diluted»
  - In methods where your final sequences are derived from one single DNA template, they become visible and must be corrected for!



Sanger



Illumina

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

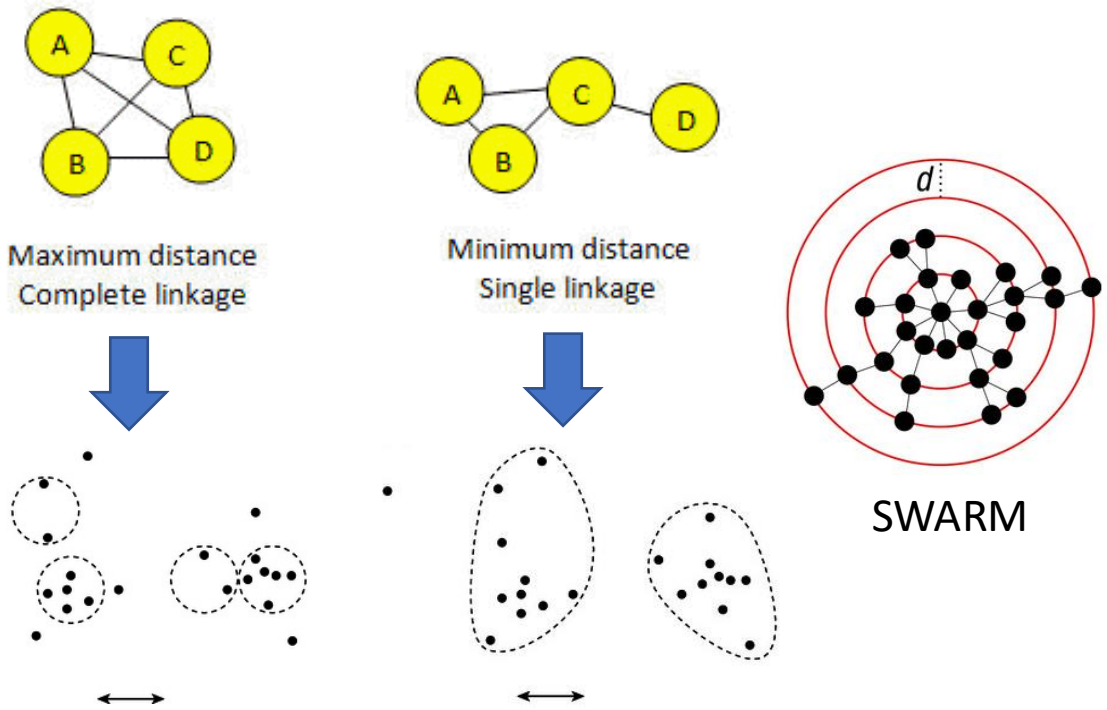
OTU modifications

Positive negatives

Singleton removal

Data transformation

Many different clustering approaches



Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

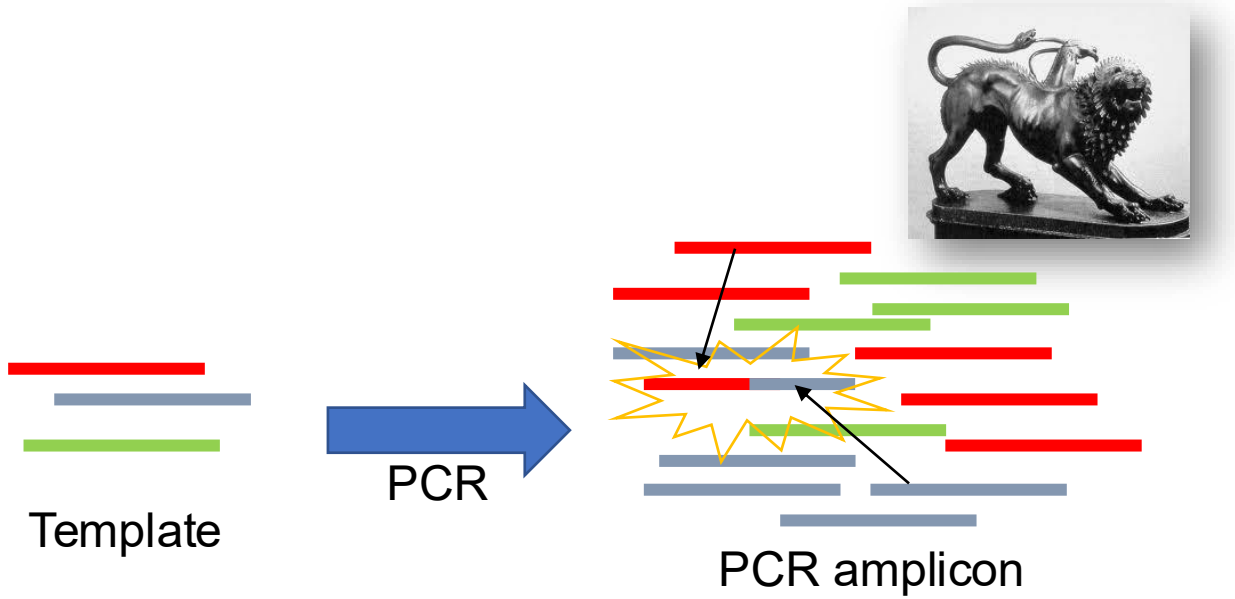
Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)



De novo versus reference based chimera checking

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

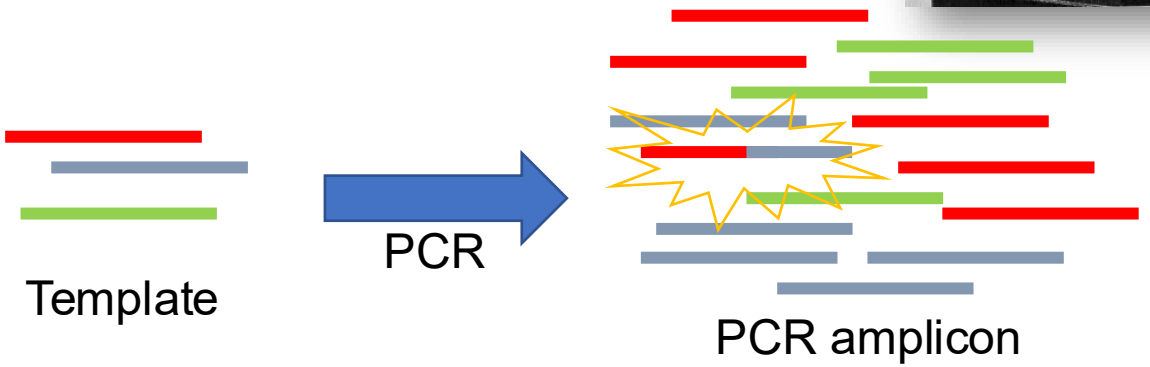
OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)





De novo ~~versus reference based~~ chimera checking

The level of chimeric sequences depends on how variable the marker is! → Be aware of false positives in the tests

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)





EUK REF





BOLD SYSTEMS

Why do taxonomic assignment?

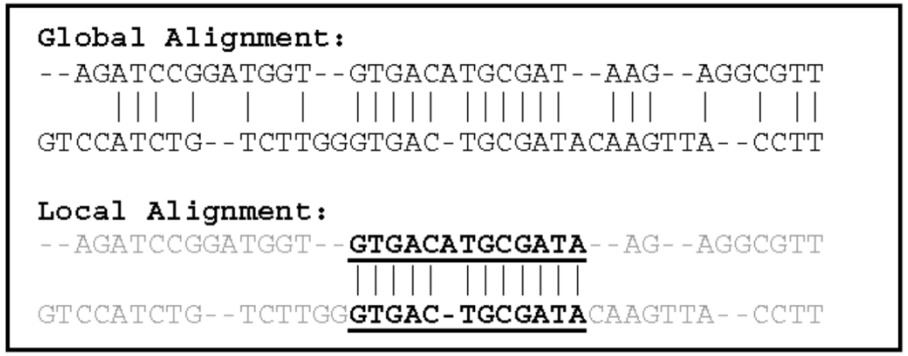
- Not strictly necessary to answer alpha and beta diversity questions
  - Detecting shifts in community composition and genetic diversity doesn't require taxonomic assignments
- Assigning taxonomy links sequences to a wealth of pre-existing information
  - Linking sequences to species improves interpretation and explanation of patterns in alpha and beta diversity
- Choice of marker can impact taxonomic assignment
  - No marker is perfect
  - Discriminating power varies between markers and taxonomic groups
  - Database quality, availability, and completeness varies between markers

Taxonomic assignment - databases

- Different algorithms for comparing sequences to databases and scoring the results
- Alignment based
  - **BLAST**, VSEARCH, OBItools
- Phylogenetic based
  - HmmUFOtu, TIPP, DECARD, SAP
- Kmer-based machine learning approaches
  - RDP, UTAX, SINTAX

Taxonomic assignment - Alignment based

- Alignment based
  - **BLAST (Basic Local Alignment Search Tool) , vsearch**
- Local alignments begin by checking a small piece of the query sequence against the reference database and then expanding the match to find areas of high similarity
- Global alignments find the best match in the reference database across the entire length of the query sequence
- Output is typically an alignment score, percent identity, and coverage score



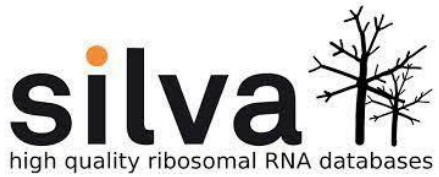
62% similarity

93% similarity
32% coverage

Taxonomic assignment - databases

- Taxonomic assignment quality is highly dependent on database accuracy and completeness
  - Misidentified sequences create identification errors and low-quality assignments
  - Missing reference sequences reduce the resolution of taxonomic assignments or result in misidentifications
- Taxonomic assignment is only as good as the database!
  - **PR2**: a curated database for protists 18S
  - **Silva**: 18S and 28S, Bacteria, Archaea, and Eukaryota. Comprehensive, but not curated.
  - **UNITE:** ITS. Mostly used for Fungi, other eukaryotes are also included
  - **ROD:** Ribosomal Operon Database, the full rDNA operon from genome projects
- Quality of taxonomic assignments can often be improved by creating custom-curated reference databases







Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

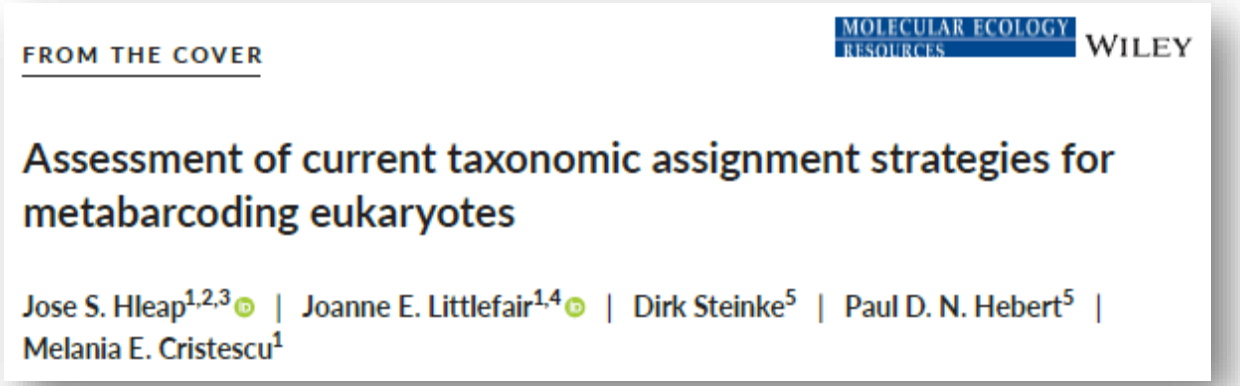
Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation (e.g rarification)



Hleap et al. 2021. Mol Ecol Resources

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

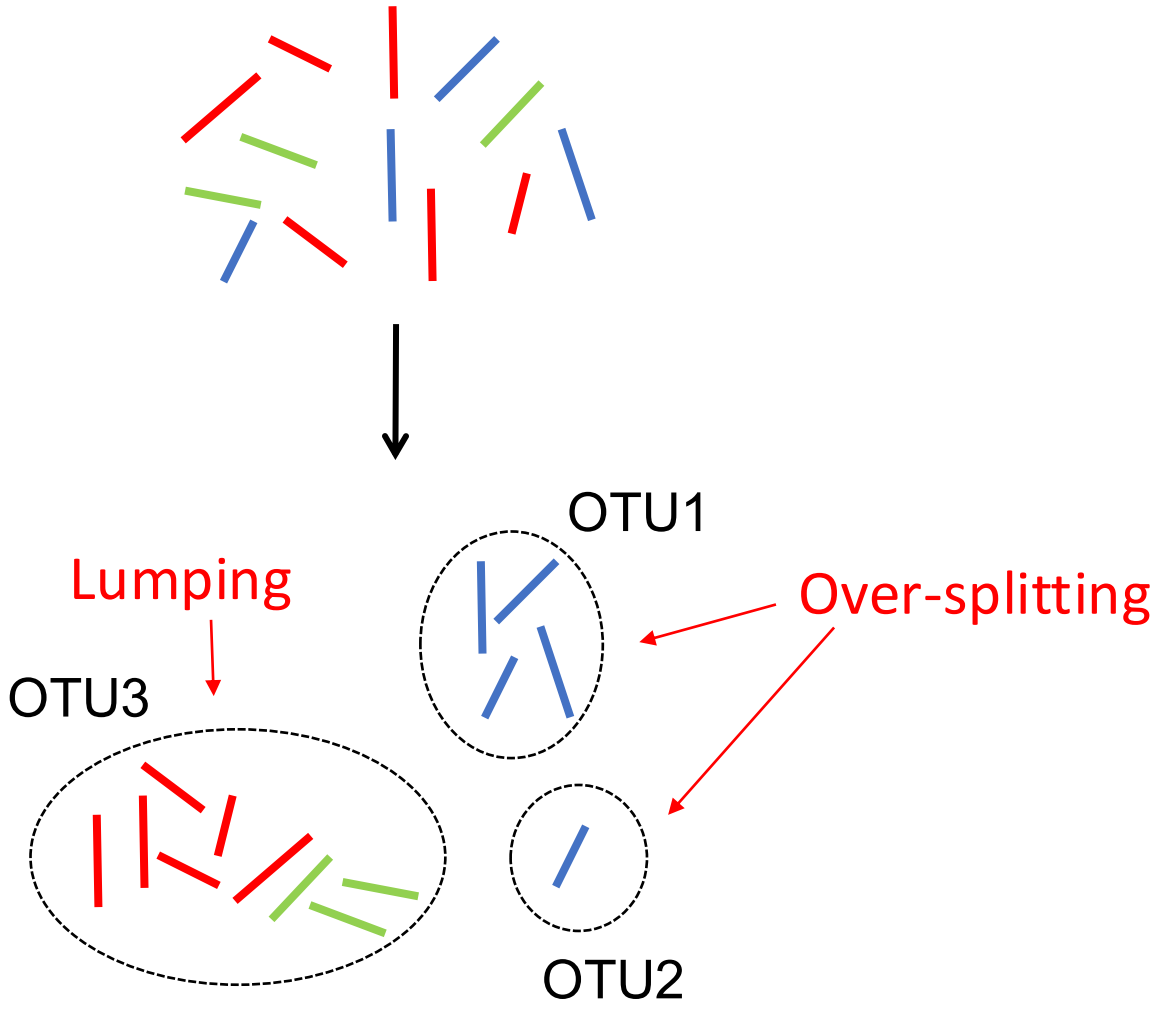
Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

Data transformation



Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

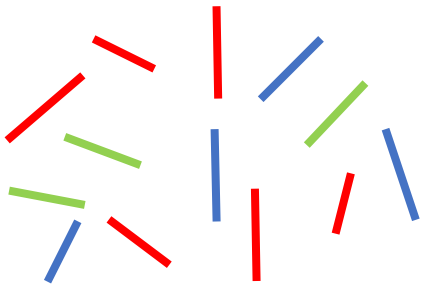
Cleaning of tag bleeding

OTU modifications

Positive negatives

Singleton removal

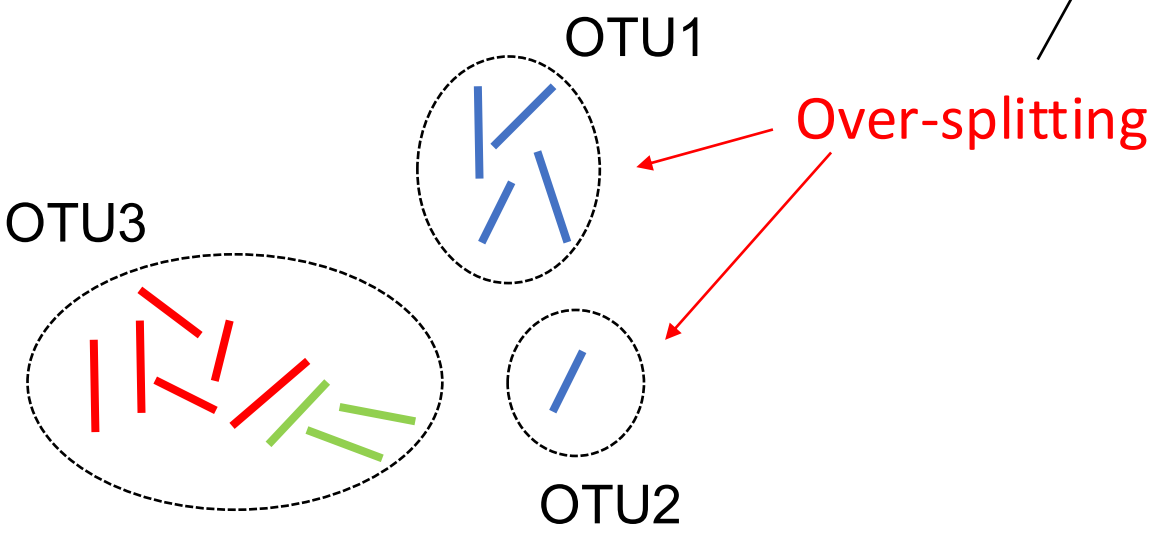
Data transformation



LULU

Samples

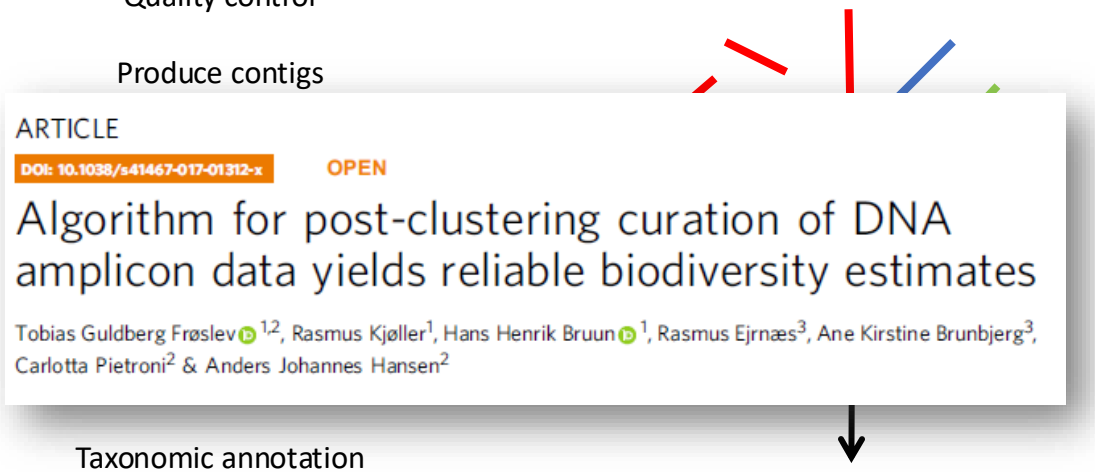
	1	2	3	4	5	6	7
OTU1	0	0	0	0	0	0	0
OTU2	2	0	10000	0	0	5	0
OTU3	0	0	0	0	0	0	0
OTU4	0	0	0	0	0	0	0
OTU5	0	0	0	0	0	0	0
OTU6	0	500	0	0	0	4	0
OTU7	0	0	0	0	0	0	0
OTU8	0	0	0	0	0	0	0
OTU9	0	0	23		0	30000	0
OTU10	0	0	0	0	0	0	0



Bioinformatics – main steps

Quality control

Produce contigs



Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

OTU modifications

Positive negatives

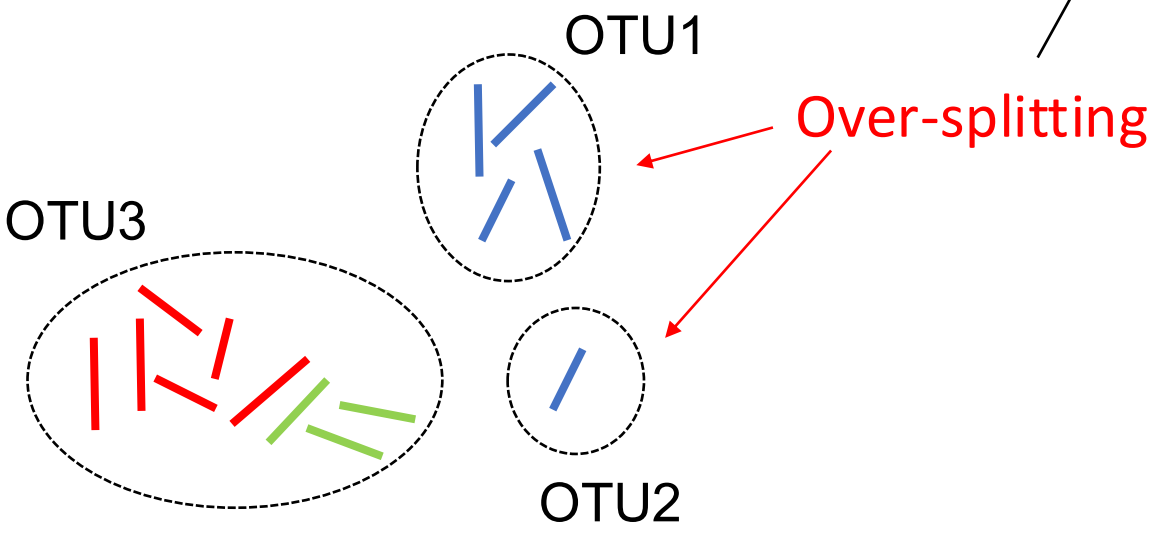
Singleton removal

Data transformation

LULU

Samples

	1	2	3	4	5	6	7
OTU1	0	0	0	0	0	0	0
OTU2	2	0	10000	0	0	5	0
OTU3	0	0	0	0	0	0	0
OTU4	0	0	0	0	0	0	0
OTU5	0	0	0	0	0	0	0
OTU6	0	500	0	0	0	4	0
OTU7	0	0	0	0	0	0	0
OTU8	0	0	0	0	0	0	0
OTU9	0	0	23		0	30000	0
OTU10	0	0	0	0	0	0	0



Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

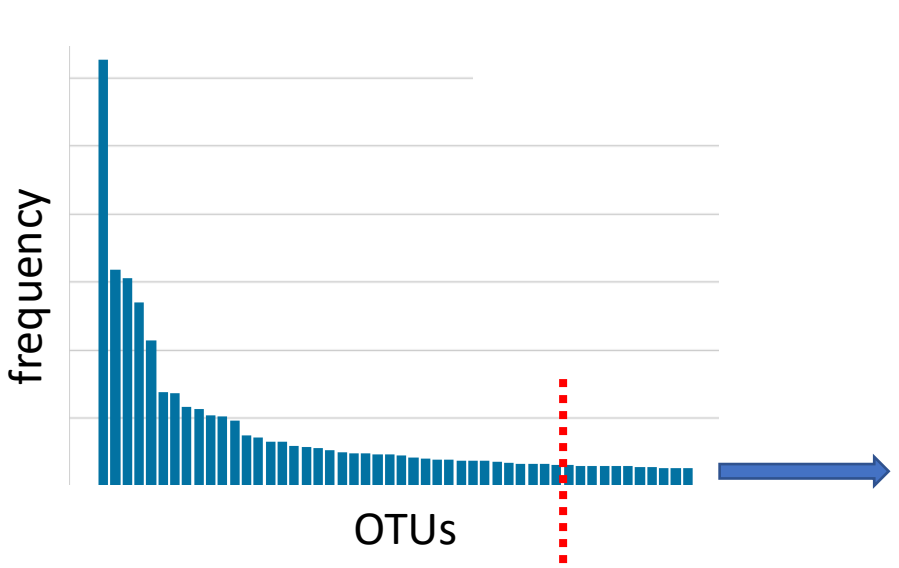
Cleaning of tag bleeding

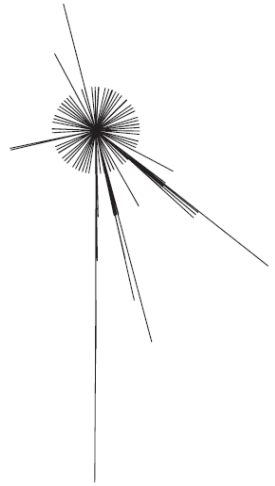
OTU modifications

Positive negatives

Singleton removal

Data transformation





What is a ‘singleton’? → Depends on the sequencing depth and quality of your data. Should also take the study aim into consideration

Bioinformatics – main steps

Quality control

Produce contigs

Demultiplexing

Dereplication

OTU construction

Chimera checking

Taxonomic annotation

Removal of non-target organisms

Cleaning of tag bleeding

OTU modifications

Positive negatives

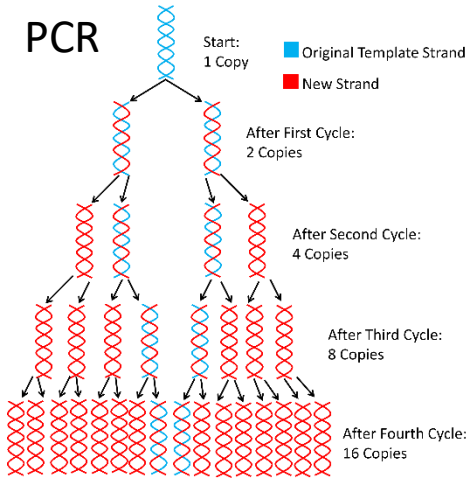
Singleton removal

Data transformation

Samples

	1	2	3	4	5	6	7
OTU1	0	0	0	0	0	0	0
OTU2	2	0	10000	0	0	5	0
OTU3	0	0	0	0	0	0	0
OTU4	0	0	0	0	0	0	0
OTU5	0	0	0	0	0	0	0
OTU6	0	500	0	0	0	4	0
OTU7	0	0	0	0	0	0	0
OTU8	0	0	0	0	0	0	0
OTU9	0	0	23		0	30000	0
OTU10	0	0	0	0	0	0	0

PCR



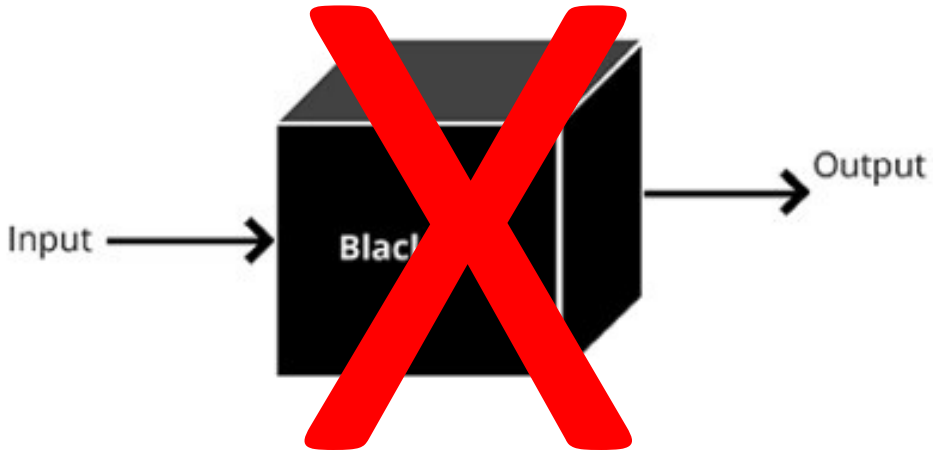
Be careful with resampling and transformations!

Check the effect from various data treatments options on the results!

Depends on the study aims!

Which methods to use???

- Which methods to use? → No general answer – it is context-dependent. You must argue for your choices!



Which methods to use?

- There are pipelines that will help with all the steps
- Qiime, LotuS
- Or you can pick different tools for different parts of the workflow to better suit your data.

Table 1 | **List of commonly used tools for metabarcoding data analysis**

Name	Description and link	Refs
DADA2	Amplicon sequence variant analysis pipeline • https://benjjneb.github.io/dada2/	38
Galaxy	Web-based platform, including various analytical tools • https://usegalaxy.org/	183
LotuS	Full pipeline for amplicon data • http://psbweb05.psb.ugent.be/lotus/index.html	47
mothur	Versatile software suite (designed mostly for 16S rRNA) • https://www.mothur.org	35
AMPtk	Full pipeline for amplicon data • http://amptk.readthedocs.io	27
OBITools	Versatile software package • https://git.metabarcoding.org/obitools	184
PipeCraft	Full pipeline for amplicon data (with graphical user interface) • https://plutof.ut.ee/#/datacite/10.15156%2FBIO%2F587450	46
PIPITS	Full pipeline for fungal ITS amplicon data (only for Illumina data) • https://github.com/hsgweon/pipits	48
QIIME	Full pipeline for amplicon data (designed mostly for 16S rRNA) • https://qiime2.org	185
SEED2	Full pipeline for amplicon data (with graphical user interface; on Windows) • http://www.biomed.cas.cz/mbu/lbwrf/seed	186
Microbiology.se	Tools, including ITSx and Metaxa2, for processing ITS, SSU and LSU data • http://microbiology.se	32,187
USEARCH	Versatile software package • https://www.drive5.com/usearch	33
VSEARCH	Versatile software package • https://github.com/torognes/vsearch	34

Nilsson et al. 2019

Biases and sources of errors

- **Extraction bias.** No universal DNA or RNA extraction method will work for all organisms and environments.
- **Marker bias**. DNA and RNA markers differ substantially in length, taxonomic resolution, copy number, and alignability.
- **Primer bias.** Primers differ in their melting temperature, binding specificity, and coverage of the targeted organisms.
- **PCR bias.** Differential amplification, mistakes in nucleotide incorporation, and chimaera formation; *keep amplification cycles low.*
- **Sequencing bias.** Differences depend strongly on the platform and sequencing strategy.
- **Bioinformatics biases.** Incorrect classification, false positive, or invalid operational taxonomic units (OTUs) formation
- **Poor clustering**. Accumulation of sequencing errors leads to wrongly constructed OTUs, which can lead to overestimation of richness. Use stringent denoising and removal of rare OTUs
- **Unequal sequencing depth**. Having different numbers of sequences between samples complicates tests of taxonomic diversity because OTU accumulation curves fail to reach a plateau in large composite samples.

Nilsson et al. 2019